

ENSEMBLE

Workforce Intelligence Platform

Product Requirements Document · v0.2 Updated Draft

Product Area	Talent Technology
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Status	Draft — For Review
Version	v0.2 — Updated Q1 2026
Last Updated	Q1 2026
Stakeholders	HRBPs, People Analytics, Talent Acquisition, Engineering, Legal

1. Problem Statement

Netflix operates a workforce unlike any other media company — four structurally distinct talent populations (Corporate FTE, Studio FTE, Union Crew, Freelance) with incompatible contract types, compensation structures, review cadences, and compliance constraints. Despite this complexity, the organization lacks a unified intelligence layer capable of surfacing risk, identifying patterns, and driving action across all four segments simultaneously.

The result is a fragmented HRBP experience: flight risk is identified reactively — often after an exit conversation — compensation gaps surface in annual reviews rather than real-time, and development interventions are uncoordinated across teams. The cost shows up in voluntary attrition, production delays from crew gaps, and lost institutional knowledge.

"Most talent systems treat all employees the same. Netflix's workforce is fundamentally segmented, and our tools should reflect that."

2. Primary Users

Persona	Core Need
HRBP (Primary)	Real-time visibility into flight risk, autonomous agent actions, and intervention tracking across their employee population, without manual data pulls.
People Analytics Lead	Cross-segment trend analysis, predictive model transparency, and exportable data to inform executive reporting and workforce planning.
VP / Executive Sponsor	Board-ready summaries of workforce health, retention risk exposure, and program ROI, without needing to navigate the platform directly.

L&D; Program Manager	Skill gap visibility by segment, L&D; participation rates, and direct linkage between development activity and flight risk reduction.
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3. Goals & Success Metrics

North Star Metric

Reduction in voluntary attrition among flight-risk-flagged employees within 90 days of HRBP intervention.

This single metric captures whether Ensemble is doing its job: surfacing the right signal, to the right person, early enough to change the outcome. All other metrics are leading indicators to this one.

Strategic Goals

- Reduce involuntary flight risk attrition by surfacing leading indicators before the exit conversation
- Increase HRBP intervention velocity — from flag to first action — across all four workforce segments
- Create a closed loop between people data, development programs, and retention outcomes
- Establish Ensemble as the single source of truth for workforce intelligence at Netflix
- Enable autonomous agent actions that execute HRBP workflows without manual effort, with full audit trails

OKRs: First 12 Months

Objective	Metric	Target
Reduce flight risk attrition	Flagged employee 90-day voluntary exit rate	↓ 25% vs. baseline
Accelerate HRBP response time	Avg. days from flag to first intervention	< 5 days (from 10.8d avg)
Increase L&D; participation	% employees with active development activity	↑ to 75% (from 61%)
Improve review compliance	% reviews completed on-cycle	↑ to 92% (from 81%)
Drive platform adoption	Weekly active HRBPs	> 90% within 6 months
Agent action adoption	% of agent-recommended actions executed within 48 hrs	> 60% within 3 months

4. Feature Requirements

MVP: Phase 1 (Q2–Q3 2026)

The MVP focuses on the core HRBP workflow: identify risk, understand context, take action.

4.1 Workforce Dashboard

- Segment-aware KPI layer: headcount, flight risk rate, L&D; participation, and open critical reqs
- Segment health scorecard: 6 live metrics per segment with directional signals and QoQ deltas
- Toggleable analytics chart: Attrition Count, Flight Risk Trend, and Time-to-Fill by segment over time

4.2 Flight Risk Engine

- Model-driven risk scoring (0–100) for all employees, refreshed weekly
- Signal attribution: comp below band, L&D; inactivity, stalled internal mobility, LinkedIn activity spike, manager change, freelance contract gap
- Sortable, filterable flight risk table with segment, intervention status, and tenure filters
- Employee detail panel: comp vs. band, tenure, engagement score, risk factors, and matched development opportunities
- Intervention tracker: status (Escalated / Pending / Actioned), owner, and date — editable by HRBP

4.3 Risk & Retention Page

- Predictive risk driver ranking with impact score and affected employee count
- Retention by tenure cohort: 6 bands from 0–12 months to 10+ years
- Intervention summary: escalation rate, resolution rate, avg. time-to-first-intervention by segment

4.4 Intelligence Agent — Updated in v0.2

The Intelligence Agent has been significantly expanded from v0.1. It now operates autonomously, executing HRBP workflows end-to-end and producing a full audit log of every action taken, skipped, or flagged for manager review. Three autonomous agent workflows are live in the current build:

Agent Action	Behavior & Governance Logic	Triggered From
Send L&D; Nudges	Identifies employees with 90+ days of L&D; inactivity, drafts personalized outreach, sends autonomously. Skips CBA-restricted union employees. Flags high flight-risk employees (75%+) for manager review.	Flight Risk / L&D; Card
Mobilize Skills Pipeline	Matches internal candidates to critical skills gaps (Unreal Engine 5, AI/ML Production, Post-Production AI, Virtual Production). Flags high-risk employees requiring a retention plan before upskilling investment. Flags external sourcing needs where no internal pipeline exists.	Skills Gap Card
Activate Onboarding Sequence	Contacts hiring managers for all 43 open critical reqs. Flags departments with no documented 90-day onboarding plan. Flags HRBP/People & Culture roles for People Analytics review.	New Hire Ramp Risk Card

Audit Log — All Agent Actions

Every autonomous agent run produces a real-time audit log showing:

- Sent count — outreach delivered autonomously
- Skipped count — with reason (e.g. CBA restriction, contract gap)
- Flagged count — escalated to manager review with specific flag reason
- Timestamp, agent type, and data vintage
- Expandable per-employee email preview for full transparency

Insight Cards — HRBP-Scoped

The agent generates exactly 3 prioritized insight cards per session, scoped to HRBP-owned workflows: flight risk and retention, L&D; engagement, skills development, internal mobility, and new hire onboarding risk. Union contract negotiation and CBA management are explicitly excluded — owned by Labor Relations.

4.5 AI Query Bar

- Natural language query against live workforce dataset
- Structured response: bulleted findings referenced to specific employees and data points
- Pre-populated prompt chips for common HRBP questions

4.6 Performance & Feedback

- Review cycle completion by segment with CBA-aware constraints for Union
- Feedback health metrics: upward, peer, manager, self-assessment, and follow-up 1:1 rates
- Overdue review table with urgency triage, manager attribution, and employee detail link

4.7 Skills & Development

- Skills gap analysis by segment: % gap, affected headcount, production-blocking context
- L&D; program enrollment vs. capacity with skill gap addressed and current status
- No-activity table: employees with 6mo+ inactivity directly linked to flight risk score

Phase 2 Roadmap (Q4 2026–Q1 2027)

The following capabilities address higher-complexity data dependencies and were deferred from MVP to maintain focus on the core HRBP workflow.

Feature	Priorit y	Dependency	Value
Manager Health Dashboard	P0	HRIS + Perf data	Surfaces retention risk concentrated under specific managers
Internal Mobility Pipeline	P0	ATS integration	Closes the loop on stalled mobility — the #3 risk driver
Compensation Band Health	P1	Comp system API	Segment-level view of below-band exposure for Finance conversations
Onboarding Cohort Tracker	P1	HRIS + L&D; data	Identifies where in year one attrition occurs for targeted intervention

Feature	Priority	Dependency	Value
Exit Interview Signal Loop	P1	Survey platform	Closes the prediction loop; validates model accuracy
Notification & Action Layer	P2	Email / Slack API	Pushes risk alerts to HRBPs without requiring platform login
Mobile View	P2	Frontend only	HRBPs need to act on signals in the field, not just at a desk
Agent Scheduler	P2	Cron / Workflow	Run autonomous agent actions on a schedule without manual trigger

5. Technical Considerations & Data Dependencies

Data Sources Required: MVP

Data Source	System	Refresh Cadence	Risk if Unavailable
Employee profiles + tenure	Workday / HRIS	Daily	Flight risk model non-functional
Compensation vs. band	Comp system	Real-time	Top risk driver (#1) missing
L&D; activity + enrollment	Learning platform	Weekly	Top risk driver (#2) missing
Performance reviews	Perf system	On completion	Review compliance page non-functional
LinkedIn activity signals	LinkedIn Talent Insights API	Weekly	Degrades model accuracy ~12%
CBA terms + expiry dates	Legal / Labor Relations	On change	Union compliance context missing
Internal job applications	ATS (Greenhouse / Lever)	Real-time	Mobility signal (#3) unavailable

Architecture Notes

- All PII must comply with Netflix's data classification standards; employee-level data access governed by HRBP role permissions
- Flight risk model should be owned and interpretable by People Analytics — not a black-box vendor score
- Intelligence Agent API calls are proxied through a serverless gateway (Vercel Edge Functions in demo); no LLM API keys are present client-side or in version control
- Autonomous agent actions are logged with full audit trails before any outreach is delivered; skip and flag statuses are enforced at the prompt layer
- CBA-governed employees require a compliance layer restricting which fields are surfaced and to whom; Legal and Labor Relations review required before launch
- Segment filtering architecture must be designed for extensibility; Netflix's workforce structure may evolve without requiring platform re-engineering
- Agent context is injected per call with structured workforce data; production build will connect to live HRIS data layer

6. Data Infrastructure

This section documents the data architecture required to move Ensemble from a demo build with hardcoded context to a production system with live, governed, and secure workforce data. Intended for Engineering, People Analytics, and Legal stakeholders.

6.1 End-to-End Data Flow

Ensemble sits above existing systems of record as an intelligence and action layer. It reads from upstream HR systems, processes signals through a risk and analytics engine, and surfaces outputs to HRBPs. No data is written back to source systems in MVP.

Stage	System(s)	Cadence	What Happens
1 — Ingestion	Workday, Comp, LMS, ATS, LinkedIn API	ETL / webhook	Raw employee, comp, L&D;, and mobility data lands in Ensemble data store
2 — Normalization	Ensemble data layer	Nightly batch	Segment classification applied; CBA flags attached; PII tagged per classification policy
3 — Signal Scoring	Risk model (People Analytics owned)	Weekly refresh	Flight risk scores computed per employee using weighted signal model; stored with attribution
4 — Context Injection	Ensemble API layer	Per AI call	Relevant employee and segment data serialized into structured prompt context
5 — Agent Execution	Claude API via Vercel proxy	On HRBP trigger	Agent generates outreach or insights; governance rules applied; audit log written before delivery
6 — Audit Storage	Ensemble audit log store	On run	Every agent action — sent, skipped, flagged — written to immutable log with timestamp and HRBP ID
7 — HRBP Interface	Dashboard + Agent UI	Real-time read	Scores, signals, insight cards, and audit logs surfaced via role-gated UI

6.2 Data Classification & Access Control

Ensemble handles three tiers of employee data, each with distinct access and retention rules. Classification must be enforced at the data layer, not just the UI.

Classification	Data Fields	Who Can Access	Handling
Tier 1 — PII	Name, employee ID, comp, tenure, performance scores	HRBP for their population only; People Analytics (aggregate); no manager access	Encrypted at rest and in transit; never passed to LLM verbatim; anonymized in agent context
Tier 2 — Sensitive HR	Flight risk score, intervention status, flag reason, L&D; inactivity	HRBP and People Analytics only; excluded from executive dashboards without aggregation	Access logged; 12-month retention; right-to-access compliant
Tier 3 — Operational	Headcount, segment totals, open reqs, skills gap %	Broad HRBP, L&D;, and executive access	Standard access controls; exportable for reporting

Classification	Data Fields	Who Can Access	Handling
Union — Restricted	Any CBA-governed employee data	Labor Relations + Legal only; HRBPs see aggregate union data only	Hard block on individual-level union data in agent calls; skip logic enforced at prompt layer

6.3 Refresh Cadence & Latency SLAs

Staleness in workforce data directly affects agent quality. The table below defines target refresh cadences and degradation behavior when SLAs are missed.

Data Source	Target Cadence	SLA	Degradation Behavior
Employee profiles	Daily overnight	< 24 hrs	Show data-age indicator; suppress flight risk scoring for stale records
Compensation vs. band	Webhook real-time	< 1 hr	Fall back to last known comp; display warning badge on affected cards
L&D; activity	Weekly Monday AM	< 7 days	Display last-refreshed date; disable L&D; nudge button if data > 14 days old
Flight risk scores	Weekly Sunday night	< 7 days	Show score vintage; disable agent trigger if scores > 14 days old
ATS / open reqs	Webhook real-time	< 2 hrs	Show req count with timestamp; flag if > 6 hours stale
LinkedIn signals	Weekly	< 7 days	Degrade model gracefully; note signal gap in risk attribution panel
CBA / contract dates	On change (manual)	Same day	Alert People Analytics if CBA data not updated within 48 hrs of known change

6.4 Agent Data Pipeline

The Intelligence Agent's quality is determined entirely by the quality of context injected into each call. The demo build uses hardcoded Q1 2026 data. The production architecture replaces this with a live context serialization layer.

Component	When	What It Does	Constraint
Context builder	On agent trigger	Queries Ensemble data layer for relevant employees, segment stats, and risk signals; serializes into structured prompt context	Must complete in < 2 seconds to maintain UX responsiveness
PII sanitization	Before context injection	Replaces raw PII with role-scoped references; ensures names are only included for the HRBP's own population	Tier 1 data never passed verbatim to external LLM API
Governance rule engine	Before agent call	Applies skip/flag rules (CBA restrictions, flight risk thresholds, contract gaps) as hard constraints before prompt is sent	Rules defined in code, not prompt — cannot be overridden by model output
Proxy layer	Every API call	Routes all LLM calls through Vercel Edge Function; injects API key from encrypted env var; enforces CORS origin allowlist	No API credentials exposed client-side or in version control
Audit writer	After agent response	Parses agent JSON output; writes each action to immutable audit log with HRBP ID and timestamp	Audit log is append-only; no delete capability; retained per HR data policy
Response renderer	After audit write	Displays audit log to HRBP in real time; surfaces per-employee email previews; marks run complete	UI shown only after audit write confirmed — no silent failures

6.5 Privacy & Compliance Layer

Ensemble processes employee data subject to California CCPA, GDPR (for non-US employees), and Netflix's internal data classification policy. The following requirements must be satisfied before production launch.

- CCPA compliance: employees have the right to know what data is collected and how it is used — a privacy notice specific to Ensemble must be approved by Legal before launch
- GDPR Article 22: automated decision-making that significantly affects employees requires human review capability — flight risk scores must be presented as advisory, not determinative
- Right to access: employees may request to see their own data held by Ensemble — a DSAR workflow must be defined pre-launch
- Data minimization: only data required for the specific signal or agent function is ingested — no speculative data collection
- Retention policy: employee-level risk scores and intervention logs retained 12 months then anonymized; audit logs retained 36 months per HR compliance standards
- Union data firewall: a hard technical block — not just a policy — must prevent individual CBA-governed employee data from reaching the agent context layer without Labor Relations approval
- Vendor DPA: a Data Processing Agreement must be executed with Anthropic before any real employee data is passed through the agent pipeline — demo build uses synthetic data only

6.6 Integration Architecture

The following integrations are required for a production Ensemble deployment. Each carries distinct engineering complexity and stakeholder dependencies.

System	Data Exchanged	Method	Priority	Owner
Workday (HRIS)	Employee profiles, org structure, tenure, manager hierarchy	REST API or SFTP	P0 — MVP blocker	HRIS Eng + IT Security
Compensation System	Salary, band min/mid/max, last adjustment date	REST API (webhook preferred)	P0 — MVP blocker	Total Rewards + Eng
LMS / Learning Platform	Course completions, enrollment, last activity date	REST API	P0 — MVP blocker	L&D; + Vendor
ATS (Greenhouse / Lever)	Open reqs, internal applications, hiring stage	REST API (real-time)	P0 — MVP blocker	Talent Acquisition + Eng
Performance System	Review completion, score, last review date, manager	REST API or batch export	P0 — MVP blocker	People Ops + Eng
LinkedIn Talent Insights	Engagement and activity signals (aggregated)	LinkedIn API (weekly)	P1 — model enhancement	People Analytics + Procurement
Email / Slack	Deliver agent-generated outreach in production	SMTP or Slack API	P2 — Phase 2	Eng + IT Security
Anthropic API (Claude)	LLM inference for Intelligence Agent and Query Bar	REST via Vercel proxy	P0 — integrated in demo	Eng + Legal (DPA required)

7. Open Questions & Risks

Open Questions

- **Data access governance:** Who sees individual employee risk scores? HRBPs only, or also direct managers? This has significant trust and adoption implications.
- **Model transparency:** Does Netflix's culture of transparency require disclosure to employees that a risk model exists? Requires HR policy decision pre-launch.
- **Union constraints:** What data can be collected for CBA-governed employees without violating labor agreements? Legal + Labor Relations review required before any union data is ingested.
- **HRBP workflow integration:** Does Ensemble replace existing tools (Workday reporting, manual trackers) or sit alongside them? Scope and change management must be defined early.
- **AI accuracy threshold:** What is the acceptable false positive rate for flight risk flags before HRBPs lose trust in the model? Define before launch, not after.
- **Autonomous action governance:** What approval layer, if any, is required before agent-sent outreach goes live in production? Demo build runs fully autonomously; production may require an HRBP confirmation step.
- **Anthropic DPA:** A Data Processing Agreement with Anthropic is required before real employee data flows through the agent pipeline. Who owns this procurement — Legal or Engineering?

Risks

Risk	Severity	Mitigation
HRIS data quality is inconsistent across segments, especially freelance	High	Audit data completeness before launch; degrade gracefully with confidence indicators on affected data
HRBPs don't trust or act on AI-generated risk scores	High	Show signal attribution, not just scores; run a pilot with 2–3 HRBPs before broad rollout
Union data handling creates legal exposure	Critical	Legal review gate before any union data ingestion; CBA compliance layer must be built-in, not bolted on
Intelligence Agent outputs reference inaccurate data	Medium	Ground all AI calls with explicit data context; validate outputs against known data points in QA
Platform adoption stalls without manager workflow integration	Medium	Phase 2 notification layer is critical for habit formation; consider a manager-facing lite view in Phase 2
Autonomous agent sends outreach to wrong employees	High	Governance rules enforced at prompt layer; production build requires HRBP confirmation before first live run
Anthropic DPA not in place before launch	Critical	No real employee data should flow through the agent pipeline until DPA is signed; Legal must be engaged in parallel with engineering build

8. Appendix: What Ensemble Is Not

Ensemble is intentionally scoped to intelligence and workflow, not system of record. Netflix already has best-in-class tools for compensation, recruiting, and performance management. Ensemble's job is to sit above those systems, connect their signals, and surface what matters. The following are deliberate non-goals:

- A compensation management tool — Ensemble surfaces comp signals but does not execute comp changes
- A union negotiation or CBA management platform — that workflow lives with Labor Relations
- A performance management system — Ensemble reads review data but does not conduct or store reviews
- A recruiting or ATS tool — talent acquisition workflows are out of scope; headcount data is read-only
- A general-purpose HR chatbot — the AI query bar and Intelligence Agent are scoped to workforce intelligence only, not benefits, policy, or HR ops questions
- An email system of record — autonomous agent outreach is a workflow accelerator, not a communication platform; production integration with existing email infrastructure is required

"Ensemble's job is to make the right intervention obvious to the right HRBP at the right time. Every feature decision should be evaluated against that sentence."

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